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Wigner-Type Theorems for Hilbert Grassmannians

Mark Pankov



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Preface

Wigner's theorem [67] provides a geometric characterization of unitary and anti-unitary operators as transformations of the set of rays of a complex Hilbert space or, equivalently, rank one projections. This statement plays an important role in the mathematical foundations of quantum mechanics [11, 50, 63], since rays (rank one projections) can be identified with pure states of quantum mechanical systems. We present various types of extensions of Wigner's theorem onto Hilbert Grassmannians and their applications. Most of these results were obtained after 2000, but for completeness of the exposition we include some classic theorems closely connected to the main topic (for example, Kakutani and Mackey's result on the lattice of closed subspaces of a complex Banach space [31] and Kadison's theorem on transformations preserving the convex structure of the set of states of quantum mechanical systems [30]). We use geometric methods related to the Fundamental Theorem of Projective Geometry and results in the spirit of Chow's theorem [13].

Formally, the material is based on the first part of Varadarajan's book [63, Chapters I – IV], but we do not assume that readers are familiar with this book. So, we describe briefly all relevant facts from the mathematical foundations of quantum mechanics and refer to Birkhoff and von Neumann [6] for motivations, physical background and the detailed description. Since the requirement of readers is only knowledge of the basics of linear algebra and operator theory, the book is accessible for graduate students.

I am very grateful to Antonio Pasini for useful discussions on some parts of this book. Finally, I would like to express my sincere gratitude to the anonymous reviewer whose remarks, suggestions and corrections helped me to finish the book.

Mark Pankov

Introduction

It was first observed by Birkhoff and von Neumann [6] that the logical structure of quantum mechanics is related to the orthomodular lattice formed by closed subspaces of a complex Hilbert space. On each orthomodular lattice is defined an important class of functions called *states*; all states form a convex set whose extreme points are known as *pure states* [63, Section III.3]. Gleason's theorem [26] describes the set of states for the orthomodular lattices associated to separable complex Hilbert spaces. It says that all states can be identified with bounded self-adjoint positive operators of trace one; in particular, pure states correspond to rank one projections, i.e. rays of the Hilbert space.

The classic Wigner's theorem [67] (see also [11, 50, 63]) characterizes unitary and anti-unitary operators as symmetries of quantum mechanical systems, i.e. every bijective transformation of the set of pure states preserving the transition probability is induced by a unitary or anti-unitary operator. We refer to Chevalier [12] for a history and a brief description of the physical background (for example, it is shown how to derive the Schrödinger equation for a conservative physical system from Wigner's theorem).

In this book, readers will meet two versions of Wigner's theorem. The non-bijective version says that an arbitrary transformation of the Grassmannian formed by rays of a complex Hilbert space which preserves the angles between pairs of rays (the square of the cosine of such an angle is equal to the transition probability between the corresponding pure states) is induced by a linear or conjugate-linear isometry. On the other hand, it was observed by Uhlhorn [62] that to get a unitary or anti-unitary operator it is sufficient to require that a transformation of the Grassmannian of rays is a bijection preserving the orthogonality relation in both directions and the dimension of the Hilbert space is not less than three. Note that the non-bijective analogue of the latter statement does not hold for infinite-dimensional Hilbert spaces. Uhlhorn's theorem is a simple consequence of the Fundamental Theorem of Projective Geometry

(for this reason, the dimension of the Hilbert space is assumed to be not less than three); but it reveals the following important relation between the logical structure and the probabilistic structure of quantum mechanical systems: if the logical structure is preserved, then probabilistic structure also is preserved.

The description of bijective transformations preserving the convex structure of the set of all quantum states (the set of all bounded self-adjoint positive operators of trace one) [30] is a classic application of Uhlhorn's version of Wigner's theorem. Since pure states are characterized as extreme points of the convex set of all states, every such transformation induces a bijective transformation of the set of pure states. The latter transformation preserves the orthogonality relation in both directions (this fact is non-trivial) and we come to a unitary or anti-unitary operator.

We present Wigner type theorems for Hilbert Grassmannians. It must be pointed out that we distinguish the Grassmannians whose elements are finite-dimensional subspaces (the dual objects are the Grassmannians consisting of closed subspaces of finite codimensions) and the Grassmannians formed by closed subspaces whose dimension and codimension both are infinite. Results of such a kind were first obtained in [36] and [27, 59], where the non-bijective and Uhlhorn's versions of Wigner's theorem were extended on other Grassmannians. Molnár's theorem [36] states that transformations of Grassmannians (formed by finite-dimensional subspaces) preserving the principal angles between any pair of subspaces are induced by linear and conjugate-linear isometries (except one finite-dimensional case). We generalize this result and show that it is sufficient to assume that only some types of the principal angles are preserved. Another generalization of Molnár's theorem was proved by Gehér [25].

Györy [27] and Šemrl [59] (independently) described bijective transformations of Hilbert Grassmannians preserving the orthogonality relation in both directions. Note that a non-bijective version of this result holds only for finite-dimensional Hilbert spaces. One of the applications of the Györy–Šemrl theorem is the determination of isometries of Hilbert Grassmannians with respect to the gap metric [24].

In the case when the Grassmannian consists of closed subspaces whose dimension and codimension both are infinite, a bijective transformation preserving the orthogonality relation (in both directions) is also inclusions preserving and we show that it can be extended to an automorphism of the lattice of closed subspaces. It is well known that all automorphisms of the lattice of closed subspaces of an infinite-dimensional complex normed space are induced by linear and conjugate-linear homeomorphisms of the normed space to itself (for the finite-dimensional case this fails). This fact was established by

Kakutani and Mackey [31] as a step in the proof of the following remarkable result: every orthomodular lattice formed by all closed subspaces of an infinite-dimensional complex Banach space is the orthomodular lattice associated to a complex Hilbert space.

We also investigate compatibility preserving transformations. The compatibility relation is one of the basic concepts of quantum logic. The orthomodular lattice formed by closed subspaces of a complex Hilbert space is considered as the standard quantum logic. Elements of this lattice are identified with projections, i.e. self-adjoint idempotents in the Banach algebra of bounded operators. Two closed subspaces are compatible if and only if the corresponding projections commute. Two distinct rays are compatible only in the case when they are orthogonal. For this reason, we regard statements which describe compatible preserving transformations as Wigner type theorems.

We will use geometric methods based on properties of Grassmann graphs in the spirit of [15, 45]. So, the Fundamental Theorem of Projective Geometry, Chow's theorem [13], apartments and their orthogonal analogues will be useful tools for our investigations. We include a chapter on geometric transformations of Grassmannians associated to vector spaces of arbitrary (not necessarily finite) dimension. A large portion of the results of this chapter is new and cannot be found in [15, 44, 45].

At the end, we give a few words on applications. It was noted above that Uhlhorn's version of Wigner's theorem was exploited in determining bijective transformations preserving the convex structure of the set of all quantum states. In a similar way, we will use analogues of Wigner's theorem for Hilbert Grassmannians to study linear transformations of the real vector space of self-adjoint finite-rank operators which send projections of fixed rank to projections of the same rank [1, 57, 58] or to projections of other fixed rank [49].

1

Two Lattices

We describe briefly some basic properties of the lattice formed by all subspaces of a vector space and the orthomodular lattice consisting of all closed subspaces of a complex Hilbert space. The first lattice is investigated in classic projective geometry [3]. The second is related to the logical structure of quantum mechanical systems (we refer to [19, 63] for the details and strongly recommend the short problem book [14] as a quick introduction to the topic).

1.1 Lattices

Let X be a non-empty set with a certain relation denoted by $\leq$. The pair $(X, \leq)$ is called a *partially ordered set* if for all $x, y, z \in X$ the following three conditions hold:

- $x \leq x$;
- if $x \leq y$ and $y \leq x$, then $x = y$;
- if $x \leq y$ and $y \leq z$, then $x \leq z$.

A partially ordered set $(X, \leq)$ is said to be a *lattice* if it satisfies the following additional conditions:

- for any two elements $x, y \in X$ there is the *least upper bound* $x \vee y$, i.e. an element $z \in X$ such that $x \leq z, y \leq z$ and we have $z \leq z'$ for all $z' \in X$ satisfying $x \leq z'$ and $y \leq z'$;
- for any two elements $x, y \in X$ there is the *greatest lower bound* $x \wedge y$, i.e. an element $t \in X$ such that $t \leq x, t \leq y$ and we have $t' \leq t$ for all $t' \in X$ satisfying $t' \leq x$ and $t' \leq y$.

A lattice is called *bounded* if it contains the *least* element 0 and the *greatest* element 1 such that $0 \leq x \leq 1$ for every element x . A lattice $(X, \leq)$ is *complete*

if for every subset $Y \subset X$ there is the least upper bound $\bigvee_{y \in Y} y$ and the greatest lower bound $\bigwedge_{y \in Y} y$.

An *isomorphism* between partially ordered sets $(X, \leq)$ and $(X', \leq)$ is a bijection $f : X \rightarrow X'$ preserving the order $\leq$ in both directions, i.e. for $x, y \in X$ we have

$$x \leq y \iff f(x) \leq f(y).$$

If these partially ordered sets are lattices and $f : X \rightarrow X'$ is an isomorphism between them, then

$$f(x \vee y) = f(x) \vee f(y) \text{ and } f(x \wedge y) = f(x) \wedge f(y)$$

for all $x, y \in X$; moreover, if our lattices are complete, then

$$f\left(\bigvee_{y \in Y} y\right) = \bigvee_{y \in Y} f(y) \text{ and } f\left(\bigwedge_{y \in Y} y\right) = \bigwedge_{y \in Y} f(y)$$

for any subset $Y \subset X$. Isomorphisms of bounded lattices transfer the least and greatest elements to the least and greatest elements, respectively.

A bijection $g : X \rightarrow X'$ is said to be an *anti-isomorphism* of $(X, \leq)$ to $(X', \leq)$ if it is order reversing in both directions, i.e.

$$x \leq y \iff g(y) \leq g(x)$$

for all $x, y \in X$. If our partially ordered sets are lattices and $g : X \rightarrow X'$ is an anti-isomorphism between them, then

$$g(x \vee y) = g(x) \wedge g(y) \text{ and } g(x \wedge y) = g(x) \vee g(y)$$

for all $x, y \in X$; also, we have

$$g\left(\bigvee_{y \in Y} y\right) = \bigwedge_{y \in Y} g(y) \text{ and } g\left(\bigwedge_{y \in Y} y\right) = \bigvee_{y \in Y} g(y)$$

for any subset $Y \subset X$ if our lattices are complete. Anti-isomorphisms of bounded lattices transpose the least and greatest elements.

Example 1.1 For every non-empty set X we denote by $\mathcal{L}(X)$ the set of all subsets of X . The partially ordered set $(\mathcal{L}(X), \subset)$ is a bounded lattice. If A and B are subsets of X , then their least upper bound is $A \cup B$ and their greatest lower bound is $A \cap B$. The least element of the lattice $\mathcal{L}(X)$ is the empty set and the greatest element is X . This lattice is complete. The lattices $\mathcal{L}(X)$ and $\mathcal{L}(Y)$ are isomorphic if and only if the sets X and Y are of the same cardinality. In this case, every isomorphism of these lattices is induced by a bijection between X and Y .

A bounded lattice is said to be *complemented* if for every element x there is a *complement* x' , i.e. an element x' satisfying

$$x \wedge x' = 0 \text{ and } x \vee x' = 1.$$

A *Boolean algebra* is a complemented lattice with the following distributive rules:

$$x \vee (y \wedge z) = (x \vee y) \wedge (x \vee z),$$

$$x \wedge (y \vee z) = (x \wedge y) \vee (x \wedge z).$$

Using these rules, we can show that for every element of a Boolean algebra there is the unique complement (see, for example, [63, p. 8]).

Example 1.2 The lattice $\mathcal{L}(X)$ from Example 1.1 is a Boolean algebra.

Let $(X, \leq)$ be a bounded lattice. An *orthocomplementation* is a transformation $x \rightarrow x^\perp$ such that for all $x, y \in X$ the following conditions hold:

- (1) $x \vee x^\perp = 1$ and $x \wedge x^\perp = 0$,
- (2) $x^{\perp\perp} = x$,
- (3) if $x \leq y$, then $y^\perp \leq x^\perp$.

The conditions (2) and (3) imply that the orthocomplementation is an anti-automorphism of $(X, \leq)$. Hence $0^\perp = 1$ and $1^\perp = 0$. For elements $x, y \in X$ we write $x \perp y$ and say that these elements are *orthogonal* if $x \leq y^\perp$ (this relation is symmetric, since $x \leq y^\perp$ implies that $y \leq x^\perp$).

A bounded lattice with an orthocomplementation is called *orthomodular* if for any two elements x, y satisfying $x \leq y$ we have

$$x \vee (x^\perp \wedge y) = y.$$

In such a lattice, De Morgan's laws

$$(x \vee y)^\perp = x^\perp \wedge y^\perp \text{ and } (x \wedge y)^\perp = x^\perp \vee y^\perp$$

hold true [63, Lemma 3.1]. Two elements x, y of an orthomodular lattice are said to be *compatible* if

$$x' = x \wedge (x \wedge y)^\perp \text{ and } y' = y \wedge (x \wedge y)^\perp \quad (1.1)$$

are orthogonal. For example, x and y are compatible if $x \leq y$ or $x \perp y$.

Example 1.3 Every Boolean algebra is an orthomodular lattice whose orthocomplementation is the complementation. Let x, y be elements of a Boolean algebra and let x', y' be as in (1.1). Using the distributive rules and De Morgan's laws, we establish that $x' = x \wedge y^\perp$ and $y' = y \wedge x^\perp$, which implies that